

## 29.1 Prediction error and optimism

Chapter 19 compared a short regression with a long one and found that neither wins everywhere. Leaving out regressors whose coefficients are small buys a drop in variance that can outweigh the bias it causes, and Theorem 19.1.5 located the break-even point exactly: at the design points the short fit is better iff the noncentrality Eq. 19.1.1 of the omitted block is less than the number of columns omitted. The noncentrality is unknown, so the theorem does not choose a model. It does say what a good choice should aim at, namely the error of the fitted values as predictions, and this chapter builds that into procedures.

The residual mean square describes how well a fit reproduces the data it was fitted to, and every fitting procedure bends towards those data. This section makes the size of that bend exact for any procedure, and turns it into a definition of the degrees of freedom a procedure spends.

### 29.1.1 In-sample prediction error

Throughout the chapter the data follow

$$\mathbf{Y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}, \quad E(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n, \quad (29.1.1)$$

where the mean vector  $\boldsymbol{\mu} \in \mathbb{R}^n$  is unrestricted. A candidate linear model  $\mathcal{C}(\mathbf{X})$  may or may not contain  $\boldsymbol{\mu}$ . That is the point of model selection: we want procedures whose properties do not depend on a candidate being right. A **fitting procedure** is any function  $\hat{\boldsymbol{\mu}} = \hat{\boldsymbol{\mu}}(\mathbf{Y})$  from  $\mathbb{R}^n$  to  $\mathbb{R}^n$  with  $E\|\hat{\boldsymbol{\mu}}(\mathbf{Y})\|^2 < \infty$ : least squares on a fixed model, least squares on a model chosen from the data, ridge regression, thresholding.

#### Definition 29.1.1 — Training error, prediction error, optimism

Let  $\mathbf{Y}^* = \boldsymbol{\mu} + \boldsymbol{\varepsilon}^*$ , where  $\boldsymbol{\varepsilon}^*$  is independent of  $\boldsymbol{\varepsilon}$  and has the same distribution: new responses at the same design points. For a fitting procedure  $\hat{\boldsymbol{\mu}}$ :

- (a) the **training error** is  $\overline{\text{err}} = n^{-1}\|\mathbf{Y} - \hat{\boldsymbol{\mu}}\|^2$ ;
- (b) the **in-sample prediction error** is  $\text{Err}_{\text{in}} = n^{-1} E(\|\mathbf{Y}^* - \hat{\boldsymbol{\mu}}\|^2 \mid \mathbf{Y})$ ;
- (c) the **optimism** is  $\text{Err}_{\text{in}} - \overline{\text{err}}$ , and its mean  $\omega = E(\text{Err}_{\text{in}} - \overline{\text{err}})$  is the **expected optimism**.

Expanding  $\mathbf{Y}^* - \hat{\boldsymbol{\mu}} = \boldsymbol{\varepsilon}^* + (\boldsymbol{\mu} - \hat{\boldsymbol{\mu}})$  and using the independence of  $\boldsymbol{\varepsilon}^*$  and  $\mathbf{Y}$ ,

$$\text{Err}_{\text{in}} = \sigma^2 + \frac{1}{n}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2. \quad (29.1.2)$$

So the in-sample prediction error is the irreducible  $\sigma^2$  plus the average squared error of the fitted values as estimates of the mean. Ranking procedures by  $E(\text{Err}_{\text{in}})$  is the same as ranking them by the risk  $E\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2$ , which is the criterion of Theorem 19.1.5(c). *In-sample* refers to the design points: the responses are new, the regressor values old. New regressor values are taken up at the end of the section.

## 29.1.2 The covariance penalty

### Theorem 29.1.1 — Optimism is a covariance

Under Eq. 29.1.1, let  $\hat{\boldsymbol{\mu}}$  be a fitting procedure.

- (a) The expected optimism is

$$\omega = \frac{2}{n} \sum_{i=1}^n \text{Cov}(\hat{\mu}_i, Y_i).$$

- (b) If  $\hat{\boldsymbol{\mu}} = \mathbf{S}\mathbf{Y}$  for a fixed  $n \times n$  matrix  $\mathbf{S}$ , then  $\omega = 2\sigma^2 \text{tr}(\mathbf{S})/n$  and  $\mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 = \|(\mathbf{I} - \mathbf{S})\boldsymbol{\mu}\|^2 + \sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top)$ .
- (c) For least squares on a model matrix  $\mathbf{X}$  of rank  $p$ , whether or not  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ ,

$$\begin{aligned} \mathbb{E}(\overline{\text{err}}) &= \sigma^2 \frac{n-p}{n} + \frac{\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2}{n}, \\ \mathbb{E}(\text{Err}_{\text{in}}) &= \sigma^2 \frac{n+p}{n} + \frac{\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2}{n}, \end{aligned}$$

and  $\omega = 2\sigma^2 p/n$ .

*Proof.*

- (a) Write  $\mathbf{Y} - \hat{\boldsymbol{\mu}} = (\mathbf{Y} - \boldsymbol{\mu}) - (\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})$ . Then

$$\|\mathbf{Y} - \hat{\boldsymbol{\mu}}\|^2 = \|\boldsymbol{\varepsilon}\|^2 + \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 - 2\boldsymbol{\varepsilon}^\top(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}).$$

Take expectations and compare with Eq. 29.1.2:  $n \mathbb{E}(\text{Err}_{\text{in}}) = n\sigma^2 + \mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2$ , so  $n\omega = 2 \mathbb{E}[\boldsymbol{\varepsilon}^\top(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})] = 2 \sum_i \mathbb{E}[(Y_i - \mu_i)\hat{\mu}_i]$ , since  $\boldsymbol{\mu}$  is constant and  $\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ . Because  $\mathbb{E}(Y_i - \mu_i) = 0$ , the last expectation is  $\text{Cov}(\hat{\mu}_i, Y_i)$ . The moments exist by the Cauchy–Schwarz inequality.

- (b)  $\text{Cov}(\mathbf{S}\mathbf{Y}, \mathbf{Y}) = \sigma^2 \mathbf{S}$ , whose diagonal sums to  $\sigma^2 \text{tr}(\mathbf{S})$ . For the risk,  $\mathbf{S}\mathbf{Y} - \boldsymbol{\mu} = \mathbf{S}\boldsymbol{\varepsilon} - (\mathbf{I} - \mathbf{S})\boldsymbol{\mu}$  has mean  $-(\mathbf{I} - \mathbf{S})\boldsymbol{\mu}$  and covariance  $\sigma^2 \mathbf{S}\mathbf{S}^\top$ , and Theorem 2.3.1 with  $\mathbf{A} = \mathbf{I}$  gives the sum of the squared mean and the trace.
- (c) Take  $\mathbf{S} = \mathbf{M}$ , with  $\text{tr}(\mathbf{M}) = p$  and  $\mathbf{M}\mathbf{M}^\top = \mathbf{M}$  (Proposition 6.3.4). Then  $\mathbb{E}(\text{Err}_{\text{in}}) = \sigma^2 + n^{-1}\{\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2 + \sigma^2 p\}$ , and subtracting  $\omega$  gives  $\mathbb{E}(\overline{\text{err}})$ . ■

Part (c) with  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  is Proposition 14.2.7(a). What is new is that the optimism of least squares does not depend on  $\boldsymbol{\mu}$ : a wrong model has larger training and prediction errors, but both grow by the same squared bias, so a penalty depending only on  $p$  is a sensible correction even for wrong models. Part (a) asks nothing of the procedure but finite second moments: a fit is optimistic to the extent that each fitted value follows its own response.

*Idea.* The training error of a procedure understates its prediction error by  $2/n$  times the total covariance between fitted values and responses. For least squares on  $p$  columns the covariance is  $\sigma^2 p$ , right model or wrong.

**Example. (Optimism of a wrong model)**

Take  $n = 30$  equally spaced points  $t_i$  on  $[-1, 1]$ , a curved mean  $\mu_i = 1 + t_i + 1.5t_i^2$ ,  $\sigma = 1$ , and fit by least squares a straight line in  $t$  together with two columns of pure noise, so  $p = 4$ . The model is wrong: its squared bias per observation is  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/n = 0.2054$ . Over 20000 simulated data sets the average optimism is 0.2688, against  $2\sigma^2 p/n = 0.2667$  from [Theorem 29.1.1\(c\)](#). Ridge regression ([Theorem 27.2.2](#)) on the same columns with penalty  $\lambda = 5$  is a linear smoother with  $\text{tr}(\mathbf{S}) = 3.207$ ; its simulated optimism is 0.2163, against  $2 \times 3.207/30 = 0.2138$  from part (b).

```
import numpy as np
rng = np.random.default_rng(2901)
n, sigma, reps = 30, 1.0, 20_000

t = np.linspace(-1, 1, n)
X = np.column_stack([np.ones(n), t, rng.normal(size=(n, 2))])  # line + 2
                        irrelevant columns
mu = 1 + t + 1.5 * t**2                                         # the true mean is
                        curved
p = X.shape[1]
M = X @ np.linalg.solve(X.T @ X, X.T)                          # projection onto C(
                        X)

Y = mu + sigma * rng.normal(size=(reps, n))                    # one data set per
                        row
fit = Y @ M                                                      # M is symmetric
train = np.mean((Y - fit) ** 2, axis=1)                         # training error
err_in = sigma**2 + np.mean((fit - mu) ** 2, axis=1)            # in-sample
                        prediction error
print(f"average optimism {np.mean(err_in - train):.4f}, 2 sigma^2 p / n = {2 *
      sigma**2 * p / n:.4f}")
```

[Run this code online](#)

```
lam = 5.0
S = X @ np.linalg.solve(X.T @ X + lam * np.eye(p), X.T)        # ridge smoother
fit_r = Y @ S.T
train_r = np.mean((Y - fit_r) ** 2, axis=1)
err_in_r = sigma**2 + np.mean((fit_r - mu) ** 2, axis=1)
omega_r = np.mean(err_in_r - train_r)
print(f"ridge: optimism {omega_r:.4f}, 2 sigma^2 tr(S) / n = {2 * np.trace(S) / n
      :.4f}")
```

[Run this code online](#)

### 29.1.3 Degrees of freedom of a procedure

[Theorem 29.1.1](#) suggests measuring the complexity of any procedure by the covariance it creates.

#### Definition 29.1.2 — Degrees of freedom

The **degrees of freedom** of a fitting procedure  $\hat{\boldsymbol{\mu}}$  under Eq. 29.1.1 is

$$\text{df}(\hat{\boldsymbol{\mu}}) = \frac{1}{\sigma^2} \sum_{i=1}^n \text{Cov}(\hat{\mu}_i, Y_i),$$

so that  $\omega = 2\sigma^2 \text{df}(\hat{\boldsymbol{\mu}})/n$ .

For least squares on a fixed model  $\text{df} = p$ , the number of free coefficients, and for a linear smoother  $\text{df} = \text{tr}(\mathbf{S})$ . For ridge regression with singular values  $d_j$  of  $\mathbf{X}$  this is  $\sum_j d_j^2/(d_j^2 + \lambda)$ , the effective degrees of freedom of Theorem 27.2.2, which falls from  $\text{rank}(\mathbf{X})$  to 0 as  $\lambda$  grows. If  $\text{df}$  is known,

$$\widehat{\text{Err}}_{\text{in}} = \overline{\text{err}} + \frac{2\sigma^2}{n} \text{df}(\hat{\boldsymbol{\mu}}) \quad (29.1.3)$$

is an unbiased estimate of  $E(\text{Err}_{\text{in}})$  by Theorem 29.1.1(a). Section 29.2 turns this into Mallows'  $C_p$ . For a procedure that searches, the covariance is not a count and has to be computed. Under normal errors it is computed by Stein's lemma, Lemma 27.5.1(b), proved in Section 27.5 for the James–Stein theorem: if  $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$  and each  $g_i$  is absolutely continuous in  $z_i$  with the integrability stated there, then  $E[(\mathbf{Z} - \boldsymbol{\theta})^\top \mathbf{g}(\mathbf{Z})] = \sigma^2 E \sum_i \partial g_i(\mathbf{Z})/\partial z_i$ .

### Proposition 29.1.2 — Degrees of freedom as a divergence

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ . Suppose that for each  $i$ , for almost every value of the other coordinates,  $y_i \mapsto \hat{\mu}_i(\mathbf{y})$  is absolutely continuous, and that  $E|\partial \hat{\mu}_i/\partial y_i|$  and  $E\hat{\mu}_i^2$  are finite. Then

$$\text{df}(\hat{\boldsymbol{\mu}}) = E \sum_{i=1}^n \frac{\partial \hat{\mu}_i}{\partial y_i}(\mathbf{Y}),$$

and  $\overline{\text{err}} + 2\sigma^2 n^{-1} \sum_i \partial \hat{\mu}_i/\partial y_i$  is an unbiased estimate of  $E(\text{Err}_{\text{in}})$  that does not involve  $\boldsymbol{\mu}$ .

*Proof.* Since  $E(Y_i - \mu_i) = 0$ ,  $\sum_i \text{Cov}(\hat{\mu}_i, Y_i) = E[(\mathbf{Y} - \boldsymbol{\mu})^\top \hat{\boldsymbol{\mu}}(\mathbf{Y})]$ , and each term is integrable by the Cauchy–Schwarz inequality. Lemma 27.5.1(b) with  $\mathbf{g} = \hat{\boldsymbol{\mu}}$  turns this into  $\sigma^2 E \sum_i \partial \hat{\mu}_i/\partial y_i$ . The unbiasedness statement is then Theorem 29.1.1(a). ■

The estimate in the proposition is **Stein's unbiased risk estimate** (Stein 1981). For a linear smoother the divergence is  $\text{tr}(\mathbf{S})$ , and nothing is new. The interesting cases are adaptive.

**Soft thresholding.** Take an orthonormal design,  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$ , and put  $\mathbf{Z} = \mathbf{X}^\top \mathbf{Y} \sim N_p(\mathbf{X}^\top \boldsymbol{\mu}, \sigma^2 \mathbf{I})$ . For a fit  $\hat{\boldsymbol{\mu}} = \mathbf{X}\hat{\boldsymbol{\beta}}(\mathbf{Z})$ ,  $\sum_i \text{Cov}(\hat{\mu}_i, Y_i) = \text{tr Cov}(\mathbf{X}\hat{\boldsymbol{\beta}}, \mathbf{Y}) = \text{tr Cov}(\hat{\boldsymbol{\beta}}, \mathbf{X}^\top \mathbf{Y}) = \sum_j \text{Cov}(\hat{\beta}_j, Z_j)$ , so the degrees of freedom can be computed coordinate by coordinate in the sequence model  $Z_j \sim N(\theta_j, \sigma^2)$ . The lasso in this design is soft thresholding,  $\hat{\beta}_j = \text{sign}(Z_j)(|Z_j| - \lambda)_+$  (Theorem 27.4.2). It is Lipschitz, with derivative  $1\{|Z_j| > \lambda\}$  except at  $\pm\lambda$ , so by Proposition 29.1.2 its degrees of freedom are the expected number of nonzero coefficients, as an exercise of Section 27.4 found by direct integration. The shrinkage exactly pays for the search.

**Hard thresholding.** Subset selection in the same design keeps or drops each coordinate whole,  $\hat{\beta}_j = Z_j 1\{|Z_j| > \lambda\}$  (Theorem 27.4.2 again). This jumps by  $\lambda$  at  $\pm\lambda$ , so Stein's lemma does not

apply, and the degrees of freedom exceed the expected count.

### Proposition 29.1.3 — Degrees of freedom of hard thresholding

Let  $Z \sim N(\theta, \sigma^2)$  and  $\lambda > 0$ . Then

$$\frac{\text{Cov}(Z1\{|Z| > \lambda\}, Z)}{\sigma^2} = \Pr(|Z| > \lambda) + \frac{\lambda}{\sigma} \left[ \phi\left(\frac{\lambda - \theta}{\sigma}\right) + \phi\left(\frac{\lambda + \theta}{\sigma}\right) \right],$$

where  $\phi$  is the standard normal density.

*Proof.* Write  $Z = \theta + \sigma W$  with  $W \sim N(0, 1)$ , and put  $a = (\lambda - \theta)/\sigma$ ,  $b = (-\lambda - \theta)/\sigma$ , so that  $|Z| > \lambda$  iff  $W > a$  or  $W < b$ . The covariance is  $E[\sigma W(\theta + \sigma W)1\{W > a \text{ or } W < b\}]$ . The truncated moments of the standard normal are  $E[W1\{W > a\}] = \phi(a)$ ,  $E[W1\{W < b\}] = -\phi(b)$ ,  $E[W^21\{W > a\}] = 1 - \Phi(a) + a\phi(a)$  and  $E[W^21\{W < b\}] = \Phi(b) - b\phi(b)$ , each by one integration by parts. Collecting terms,

$$\sigma\theta(\phi(a) - \phi(b)) + \sigma^2(1 - \Phi(a) + \Phi(b) + a\phi(a) - b\phi(b)).$$

Since  $\sigma\theta + \sigma^2a = \sigma\lambda$  and  $-\sigma\theta - \sigma^2b = \sigma\lambda$ , this is  $\sigma^2 \Pr(|Z| > \lambda) + \sigma\lambda(\phi(a) + \phi(b))$ , and  $\phi(b) = \phi((\lambda + \theta)/\sigma)$ . ■

The first term is the expected number of coefficients kept. The second is the price of the search: the fit follows  $Z$  most closely just where  $Z$  decides whether to keep the coordinate. At the threshold  $\lambda = \sigma\sqrt{2}$  (which [Section 29.2](#) shows is where  $C_p$  and AIC put it) and  $\theta = 0$ , the expected count is 0.157 but the degrees of freedom are 0.572: the search term 0.415 is more than twice the count. At  $\theta = \sigma$  the degrees of freedom are 0.896, and at  $\theta = 3\sigma$  they are 1.104, more than one full parameter for a single coordinate.

#### Example. (Best subset of a given size)

Keep the  $k$  largest of  $m = 20$  values  $|Z_j|$  and set the rest to zero, with  $Z_j \sim N(\theta_j, 1)$  independent. For a subset fixed in advance the degrees of freedom would be  $k$ . [Figure 29.1.1](#) shows them by simulation. When all  $\theta_j = 0$ , the best single coordinate already costs 4.91 degrees of freedom, the best five cost 13.94 and the best ten 18.27. When five means equal 6 and the rest are zero, choosing the best five is almost free (5.06), because the five are obvious; choosing a sixth means choosing among fifteen pure-noise coordinates, and the degrees of freedom jump to 9.47. Searching among many candidates of equal merit is expensive, and choosing between clear winners and clear losers is cheap.

```
m, reps_s = 20, 20_000
Zs = rng.normal(size=(reps_s, m))
```

```
def subset_df(theta, k):
    """Monte Carlo df of 'keep the k largest |Y_j|' when Y ~ N(theta, I)."""
    Yb = theta + Zs
    keep = np.argsort(-np.abs(Yb), axis=1)[: , :k]
```

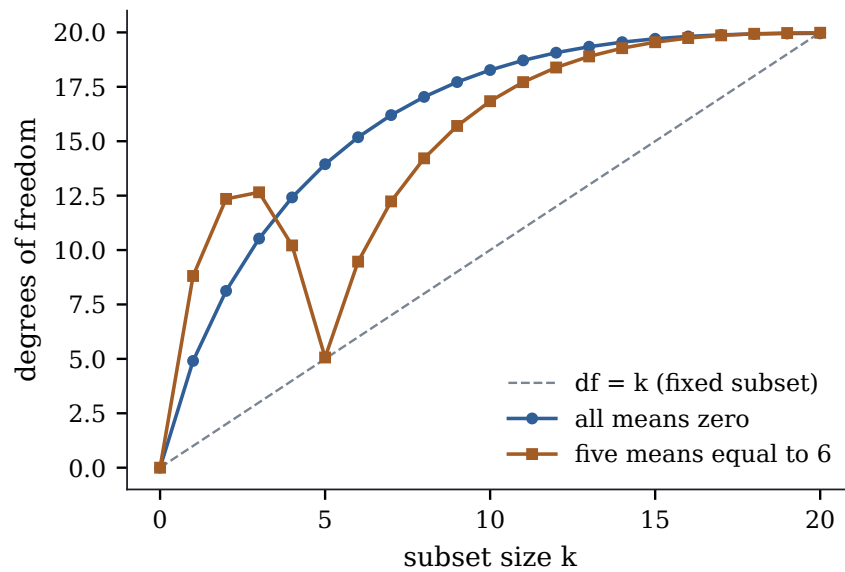


Figure 29.1.1: Degrees of freedom of the best subset of size  $k$  among 20 orthonormal coordinates, by simulation (20000 data sets). The dashed line is  $df = k$ , correct only when the subset is fixed in advance.

```

fit = np.zeros_like(Yb)
np.put_along_axis(fit, keep, np.take_along_axis(Yb, keep, axis=1), axis=1)
return np.mean(np.sum(fit * Zs, axis=1))           # sum_j Cov(fit_j,
Y_j)

null = np.zeros(m)
sparse = np.r_[np.full(5, 6.0), np.zeros(m - 5)]   # five large means
for k in (1, 5, 10):
    print(f"k = {k:2d}: df null {subset_df(null, k):5.2f}, df sparse {subset_df(
sparse, k):5.2f}")

```

[Run this code online](#)

So a penalty of  $2\sigma^2 p/n$  is right for a model chosen in advance and too small for a model chosen by search, a point [Section 29.5](#) returns to.

## 29.1.4 New regressor values

The in-sample error keeps the design fixed. For a new case with its own regressors the error is larger, and for least squares with  $k$  normal regressors and a correct model Theorem 14.2.6 gives it exactly; since  $E(s^2) = \sigma^2$ ,  $s^2(1 + 1/n)(n - 2)/(n - k - 2)$  estimates it without bias. Breiman and Spector (1992) found that fixed-design estimates understate the error when the design is random, and recommended five- or ten-fold cross-validation. Cross-validation ([Section 29.3](#)) estimates the random-design error directly.

## 29.1.5 Exercises

### 29.1.5.1 A. Check your understanding

#### Exercise A1.

The procedure  $\hat{\mu}(\mathbf{Y}) = \mathbf{Y}$  interpolates the data. Compute its training error, its in-sample prediction error and its degrees of freedom, and check [Theorem 29.1.1\(a\)](#).

### 29.1.5.2 B. Practice

#### Exercise B1.

Let  $\mathbf{S}_\lambda = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top$  be the ridge smoother, whose trace  $\sum_j d_j^2 / (d_j^2 + \lambda)$  was found in Theorem 27.2.2. Show that  $\text{tr}(\mathbf{S}_\lambda \mathbf{S}_\lambda^\top) < \text{tr}(\mathbf{S}_\lambda)$  for every  $\lambda > 0$  with  $\mathbf{X} \neq \mathbf{0}$ , and that for a symmetric smoother  $\mathbf{S}$  with eigenvalues in  $[0, 1]$ ,  $\text{tr}(\mathbf{S}\mathbf{S}^\top) = \text{tr}(\mathbf{S})$  iff  $\mathbf{S}$  is an orthogonal projection.

*Solution.* A symmetric  $\mathbf{S}$  with eigenvalues  $s_j \in [0, 1]$  has  $\text{tr}(\mathbf{S}\mathbf{S}^\top) = \text{tr}(\mathbf{S}^2) = \sum_j s_j^2 \leq \sum_j s_j = \text{tr}(\mathbf{S})$ , with equality iff every  $s_j \in \{0, 1\}$ , that is, iff  $\mathbf{S}$  is idempotent, an orthogonal projection (Theorem 6.3.3). The ridge smoother is symmetric with eigenvalues  $d_j^2 / (d_j^2 + \lambda)$ , which lie in  $(0, 1)$  for  $d_j > 0$ , and at least one  $d_j$  is positive. So the variance term of [Theorem 29.1.1\(b\)](#),  $\sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top)$ , is strictly smaller than the optimism term  $\sigma^2 \text{tr}(\mathbf{S})$ .

#### Exercise B2.

Arrange  $n$  observations on a circle, and let  $\hat{\mu}_i$  be the average of  $Y_{i-q}, \dots, Y_{i+q}$  (indices modulo  $n$ ,  $2q + 1 \leq n$ ). Find the degrees of freedom and the expected optimism, and explain why a wider window is “simpler”.

### 29.1.5.3 C. Going deeper

#### Exercise C1.

In the sequence model  $Z_j \sim N(\theta_j, \sigma^2)$ ,  $j = 1, \dots, p$ , independent, soft thresholding at  $\lambda$  has Stein’s unbiased risk estimate

$$\text{SURE}(\lambda) = \sum_{j=1}^p \min(Z_j^2, \lambda^2) - p\sigma^2 + 2\sigma^2 \#\{j : |Z_j| > \lambda\}$$

by [Proposition 29.1.2](#), because  $Z_j - \hat{\theta}_j$  is  $Z_j$  when  $|Z_j| \leq \lambda$  and  $\pm\lambda$  otherwise. Let  $0 = a_0 \leq a_1 \leq \dots \leq a_p$  be 0 and the ordered  $|Z_j|$ . Show that SURE is nondecreasing on each interval  $[a_k, a_{k+1})$ , so it is minimized over  $\lambda \geq 0$  at one of  $a_0, \dots, a_p$ . Compute  $\text{SURE}(0)$ , and explain why the minimized value is biased downwards. (Minimizing SURE is the SureShrink rule of Donoho and Johnstone 1995.)

*Solution.* On  $[a_k, a_{k+1})$  exactly  $p - k$  of the  $|Z_j|$  exceed  $\lambda$  (for distinct values), so  $\text{SURE}(\lambda) = \sum_{|Z_j| \leq a_k} Z_j^2 + (p - k)\lambda^2 - p\sigma^2 + 2\sigma^2(p - k)$ , which is nondecreasing in  $\lambda$  and smallest at  $a_k$ . At  $\lambda = 0$  nothing is shrunk and  $\text{SURE}(0) = p\sigma^2$ , the exact risk of  $\mathbf{Z}$ . Each  $\text{SURE}(a_k)$  is unbiased for its own risk, so their minimum has expectation at most the smallest risk: the optimism of a minimized estimate studied in [Section 29.5](#).

### Exercise C2.

In [Example \(Best subset of a given size\)](#) with all  $\theta_j = 0$  and  $k = 1$ , show that the degrees of freedom equal  $E \max_j Z_j^2$  (with  $\sigma = 1$ ). Deduce that they are at least 1, and use  $E \max_j Z_j^2 \geq (E \max_j |Z_j|)^2$  to explain why they grow without bound as  $m \rightarrow \infty$ .